

Generation Interconnection Request System Impact Study Report

For Queue Project Q22-1998

New Hollandale 230 kV
December 02, 2023

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q22-1998. The Interconnection Customer proposes a 191.2 MW solar plus storage facility interconnecting at the New Hollandale 230 kV substation in the Ohio area. The request was received on February 06, 2022. The study evaluated system intact and single contingency conditions using the regional planning case, including 513 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 1 overloaded facility. The total estimated cost of the network upgrades allocated to this project is \$404,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Osawatomie 100-1 kV Autotransformer	Bodo to Bluestem 230 kV	111.4	131	5.9

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Osawatomie 100 kV/1 kV Transformer	Replace transformer with higher-rated unit	\$2,511,000	16% (\$404,000)
	Total allocated to project		\$404,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.